

Zander Carpio

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Mathematics undergraduate working on quantitative modelling: epidemic dynamics, geospatial disease surveillance, and computational number theory. Seeking actuarial and data science roles.

Education

BS Mathematics Aug 2023 to 2027 (expected)

University of the Philippines Baguio

DOST-SEI Undergraduate Scholar, a competitive national merit scholarship from the Philippine Department of Science and Technology. Undergraduate thesis in number theory.

Relevant coursework: Real Analysis, Complex Analysis, Analytic Number Theory, Operations Research, Physics, Accounting.

Secondary Education 2017 to 2023

Aurora National Science High School

Experience

Intern, Statistical Operations and Coordination Division Jun to Jul 2026

Philippine Statistics Authority, Baguio City

- Statistical operations, data visualisation and coordination work inside the national statistics agency responsible for official statistics, the census and civil registration.
- Worked with survey and administrative data in its delivered state, including inconsistent labelling and failed joins.

Peer Mentor, Learning Resource Center Sep 2025 to present

University of the Philippines Baguio, Contract

Mentoring undergraduates through mathematics coursework.

Mathematics Tutor Jul 2021 to Jul 2024

Freelance, Hybrid

Three years tutoring mathematics at secondary and lower undergraduate level.

Social Media Manager Jul 2022 to present

Freelance, Remote

Content production and campaign management for small businesses, run alongside full-time study.

Quantitative Projects

Dengue surveillance analysis, Baguio City R, QGIS

Reconstructed barangay-level surveillance across 127 barangays and 23 reporting weeks, joined on PSGC codes and mapped in QGIS, resolving an outbreak that peaked at 1,778 reported cases and fell below 70 within six weeks. Found two defects that silently corrupt naive analysis of the source files: counts mislabelled as rates, and delimiter collisions inside area names.

Compartmental epidemic models R, deSolve, Shiny, JavaScript

SIR and SEIR systems integrated with **deSolve** and published as interactive Shiny applications, then reimplemented independently in JavaScript with a fourth-order Runge-Kutta integrator. The two agree on peak infected, peak timing and attack rate.

Cholera transmission with an environmental reservoir R, MATLAB

SEIRB model coupling direct transmission to a bacterial reservoir, built as a 797-line Shiny application with fourteen tunable parameters and cross-implemented in MATLAB.

Kaprekar triples, undergraduate thesis PARI/GP, LaTeX

Extending Iannucci and Foster (2005). Reduced an infinite search to a finite one by proving every triple is a root of a congruence, bounding the count at four per power of ten. Worked to a rule that no result counts until two independently written programs agree, which exposed a counting error in earlier code.

Technical Skills

Languages R, Python, MATLAB, C, JavaScript, PARI/GP

Libraries deSolve, Shiny, ggplot2, dplyr, tidyr, NumPy

Tools QGIS, LaTeX, Git, Excel